Def^n A (real) inner product space is a $\mathbb{R}$-space V over $\mathbb{R}$ equipped with an inner product i.e. a function $V \times V \xrightarrow{\langle , \rangle} \mathbb{R}$

Ref LADR

that is ① bilinear $\langle av+v', w\rangle = a\langle v, w\rangle + \langle v', w\rangle$, similar in slot 2

② symmetric: $\langle v, w\rangle = \langle w, v\rangle$

③ positive definite: $\langle v, v\rangle > 0$ for $v \neq 0$

Consequece of ③: (Rmk)
$\langle v, -\rangle = 0$ function i.e $\langle v, x\rangle = 0 \ \forall x \in V$
$\Rightarrow v = 0$ This property is called nondegeneracy of $\langle , \rangle$

Forget this ↓

~~For any $\mathbb{R}$-space V over any F~~
~~a bilinear map $V \times V \to F$ is~~
~~called a bilinear form on V~~
(Artin's context)

Examples ⓪ $\mathbb{R}^n$ with $\langle , \rangle$ = dot product

① simple but useful: any subspace of an inner product space

② $C[-1,1] \ni f, g$ $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$ (or even $\int_{-1}^{1} f(x)g(x)w(x)\,dx$ $w(x)$ positive)

$\cup$
all polynomials

Basic def^n • $\|v\| := \sqrt{\langle v, v\rangle}$ • $v \perp w = \langle v, w\rangle = 0$

Check this is valid for dot product in $\mathbb{R}^2, \mathbb{R}^3$

• For $v, w \neq 0$ $\exists!\ \theta \in [0, \pi]$ with $\cos\theta = \dfrac{\langle v, w\rangle}{\|v\|\,\|w\|}$

b/c <u>Claim</u> (Cauchy-Schwarz $|\langle v, w\rangle| \leq \|v\|\,\|w\|$ with equality when .... <u>Pf later</u>

• A basis $e_1, e_2, \ldots, e_n$ of Euclidean V is called orthonormal (O.N.) if

$\langle e_i, e_i\rangle = 1$, $\langle e_i, e_j\rangle = 0$ for $i \neq j$

<u>Example</u>: Std basis for $(\mathbb{R}^n$, dot product)

<u>Exercise</u>: for polynomials of deg $\leq n$?

• It is easy to find coord of a vector wrt an O.N basis: If $v = c_1 e_1 + \cdots + c_n e_n$

then $\langle e_j, v\rangle = \langle e_j, \sum_i c_i e_i\rangle$ gives $c_j = \langle e_j, v\rangle$

$\therefore$ $\boxed{v = \sum_j \langle e_j, v\rangle e_j}$

<u>Rmk</u>: Any set of mutually orthogonal vectors is linearly indep. Let $w_1, \ldots, w_k$ be such vectors

Let $\sum_i c_i w_i = 0$ Apply $\langle w_i, -\rangle$ to get $c_i \langle w_i, w_i\rangle = 0$

Def^n Let W be a subspace of V. Define the "orthogonal complement of W":

$$W^\perp := \{ v \in V \mid v \perp W \text{ i.e } \langle v, w \rangle = 0 \ \forall w \in W \}$$ Note $W^\perp$ is a subspace of V

We will see another proof via explicit formulas using the Gram-Schmidt process

Thm $V = W \oplus W^\perp$ **Pf** $W \cap W^\perp = 0$ b/c for w in the intersection $\langle w, w \rangle = 0$,

Let $\dim V = n$ $\dim W = k$ We claim $\dim W^\perp \geq n-k$. Why?

Take basis $w_1, \dots, w_k$ for W. Then $W^\perp = \ker\langle w_1, - \rangle \cap \ker\langle w_2, - \rangle \cap \cdots \cap \ker\langle w_k, - \rangle$

$N_1 = \ker\langle w_1, - \rangle$ has $\dim \geq n-1$

$N_2 = \ker\langle w_2, - \rangle|_{N_1}$ has $\dim \geq n-2$

and so on. $N_k = W^\perp$, $\dim N_k \geq n-k$

Now $n \geq \dim(W + W^\perp) = \dim W + \dim W^\perp - \dim(W \cap W^\perp)$

$\geq k + n-k = n$ $\therefore$ $\boxed{V = W \oplus W^\perp}$

$\therefore$ Must have equality throughout

Class next Tue @ 9-10:15 pm

Scale all e_i to have norm 1.

Consequences: ① There are lots of O.N. bases

Take $e_1 \neq 0$, $e_2 \in e_1^{\perp} \to \dim n-1$, $e_3 \in \{e_1, e_2\}^{\perp} \to \dim n-2$, ... (or induction)

② Def^n Given subspace $W \subset V$, $v \in V$, we can write $v = w_1 + w_2$ uniquely with $w_1 \in W$, $w_2 \in W^{\perp}$

Define $P(v) = w_1$ P = orthogonal projection onto W

(P is clearly linear and $P^2 = P$)

③ $(W^{\perp})^{\perp} = W$ b/c $\begin{cases} \dim (W^{\perp})^{\perp} = \dim W \\ (W^{\perp})^{\perp} \supset W \end{cases}$

Exercises

① Fix nonzero $v \in$ inner product space V. Find formula for projection onto $\mathbb{R}v$ i.e. u = (component of u along v) + (component $\perp$ to v)

② Give two proofs of Cauchy-Schwarz
- reduce to dot product
- using previous exercise

③ Find P such that $P^{-1}\begin{bmatrix} a & b \\ b & d \end{bmatrix}$ P is diagonal